



**FPT SCHOOL OF BUSINESS
& TECHNOLOGY**

FAST FOURIER TRANSFORM

OBJECTIVES

- Understand the FFT algorithm.
- Know the computation of the discrete Fourier transform

1. Decimation-in-time FFT Algorithms

2. Decimation-in-frequency FFT Algorithms

- Makes use of both periodicity and symmetry
- Consider the special case of N an integer power of 2
- Separate $x[n]$ into two sequence of length $N/2$
 - Even indexed samples in the first sequence
 - Odd indexed samples in the other sequence

$$\begin{aligned}
 X[k] &= \sum_{n=0}^{N-1} x[n] e^{-j(2\pi/N)kn} \\
 &= \sum_{n \text{ even}} x[n] e^{-j(2\pi/N)kn} + \sum_{n \text{ odd}} x[n] e^{-j(2\pi/N)kn}
 \end{aligned}$$

- Substitute variables $n=2r$ for n even and $n=2r+1$ for odd

$$X[k] = \sum_{n \text{ even}} x[n] e^{-j(2\pi/N)kn} + \sum_{n \text{ odd}} x[n] e^{-j(2\pi/N)kn}$$

$$\begin{aligned} X[k] &= \sum_{r=0}^{N/2-1} x[2r] W_N^{2rk} + \sum_{r=0}^{N/2-1} x[2r+1] W_N^{(2r+1)k} \\ &= \sum_{r=0}^{N/2-1} x[2r] W_{N/2}^{rk} + W_N^k \sum_{r=0}^{N/2-1} x[2r+1] W_{N/2}^{rk} = G[k] + W_N^k H[k] \end{aligned}$$

$$W_N^2 = e^{-j\frac{2\pi}{N}2} = e^{-j\frac{2\pi}{N/2}} = W_{N/2}$$

- $G[k]$ and $H[k]$ are the $N/2$ -point DFT's of each subsequence

$$\begin{aligned}
 X[k] &= \sum_{r=0}^{N/2-1} x[2r] W_{N/2}^{rk} + W_N^k \sum_{r=0}^{N/2-1} x[2r+1] W_{N/2}^{rk} \\
 &= G[k] + W_N^k H[k] \\
 e^{-j\frac{2\pi}{N} 2rk} &= e^{-j\frac{2\pi}{N/2} rk} = W_{N/2}^{rk}
 \end{aligned}$$

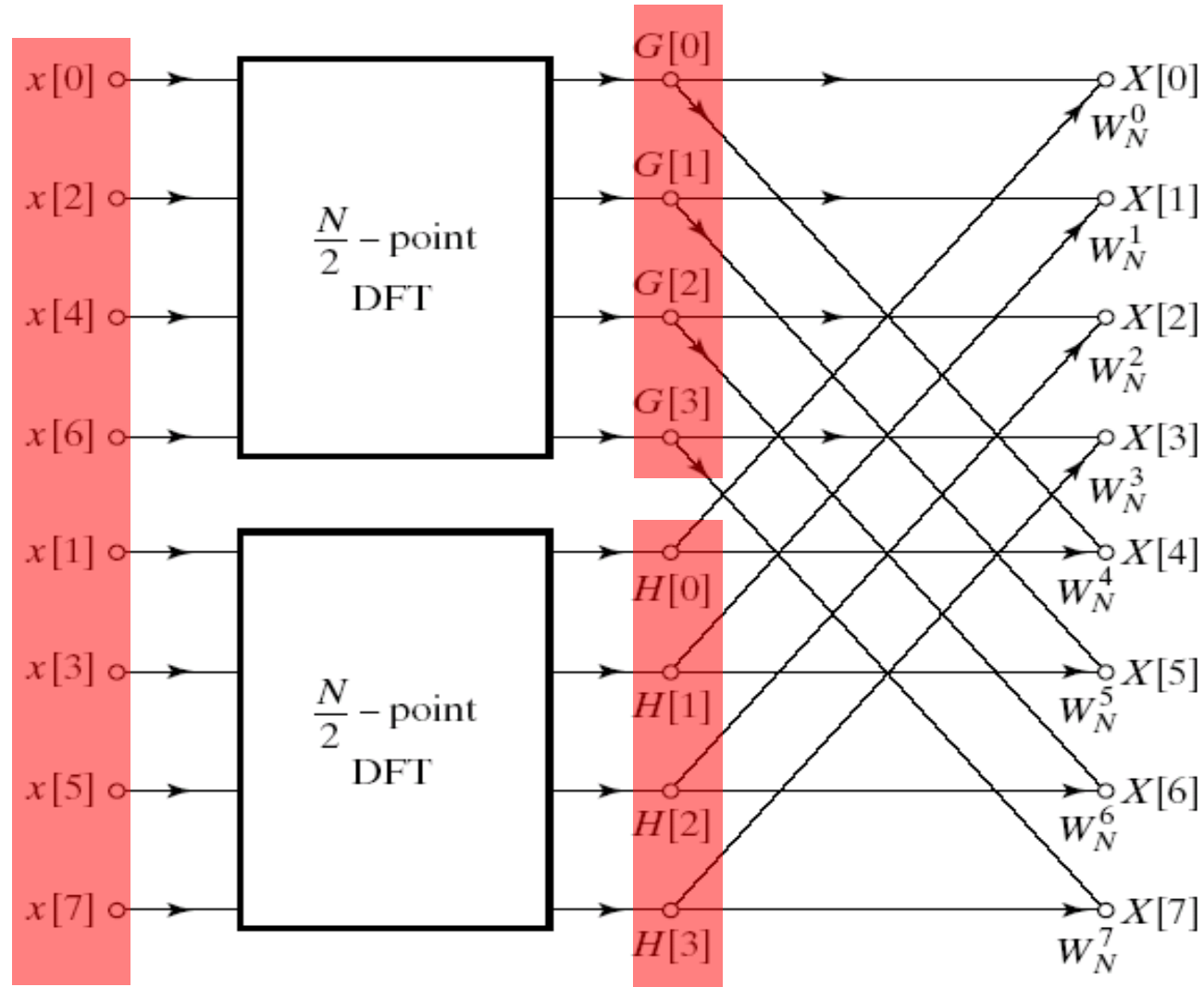
$$k = 0, 1, \dots, \frac{N-1}{2} \quad \longrightarrow \quad k = 0, 1, \dots, N$$

$$G\left[k + \frac{N}{2}\right] = G[k]$$

$$H\left[k + \frac{N}{2}\right] = H[k]$$

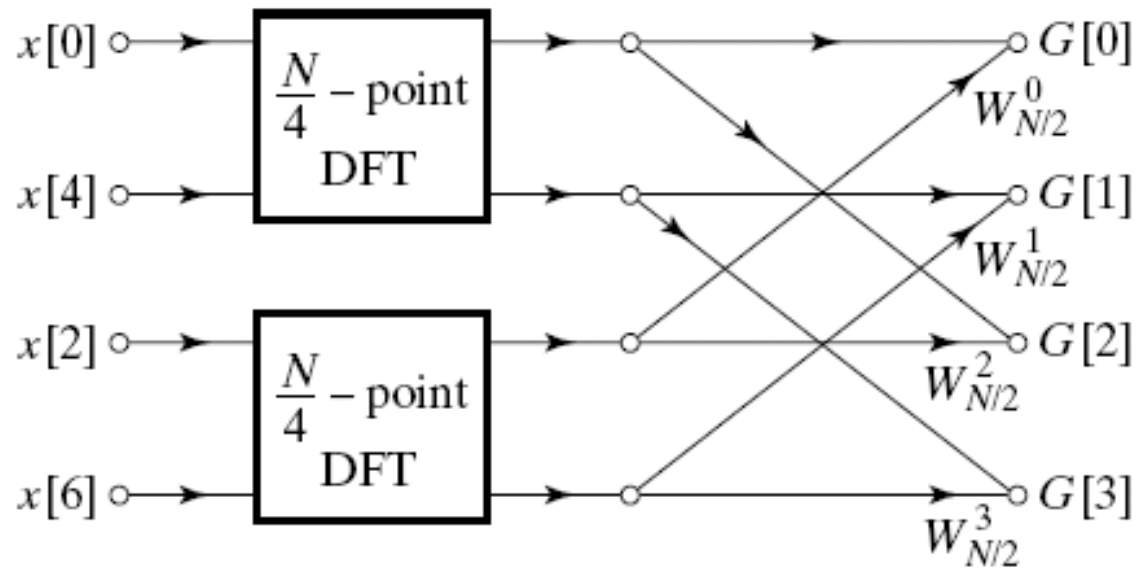
- $G[k]$ and $H[k]$ are the $N/2$ -point DFT's of each subsequence
- 8-point DFT using decimation-in-time

DECIMATION-IN-TIME FFT ALGORITHMS

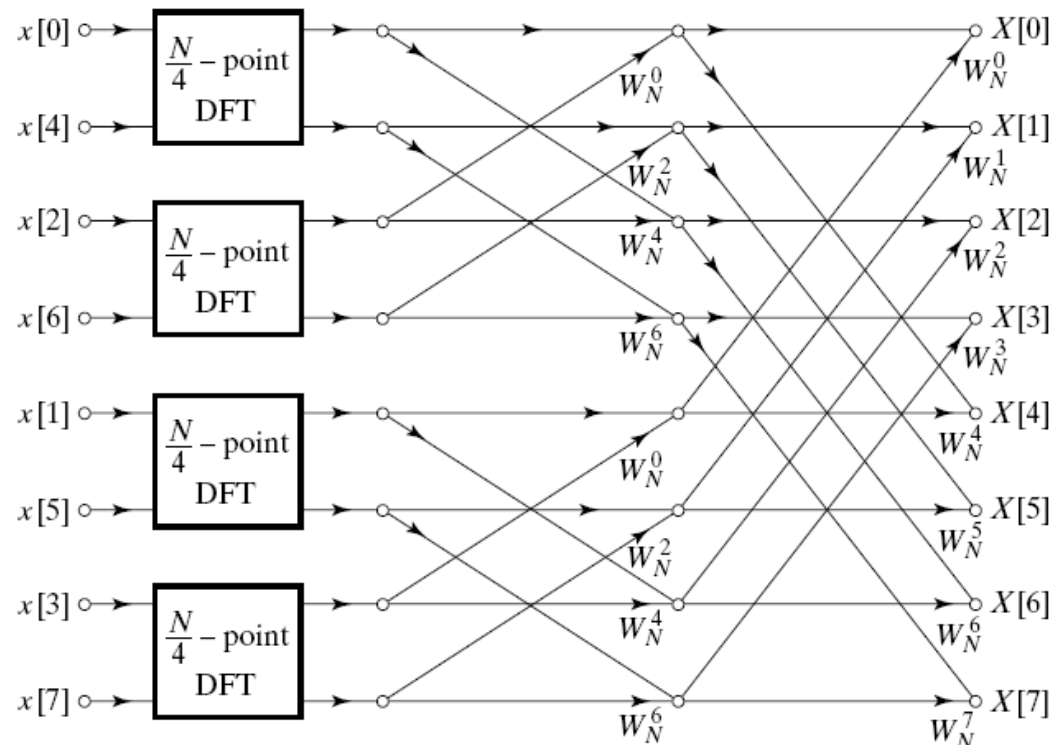


- Computational complexity
 - Two $N/2$ -point DFTs
 - $2(N/2)^2$ complex multiplications
 - $2(N/2)^2$ complex additions
 - Combining the DFT outputs
 - N complex multiplications
 - N complex additions
 - Total complexity
 - $N^2/2 + N$ complex multiplications
 - $N^2/2 + N$ complex additions

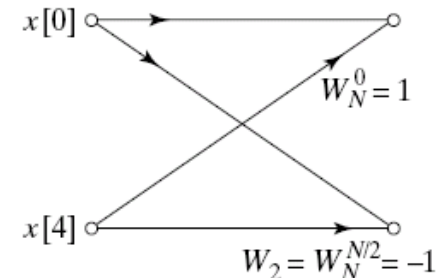
- Decimation-in-time FFT Algorithms



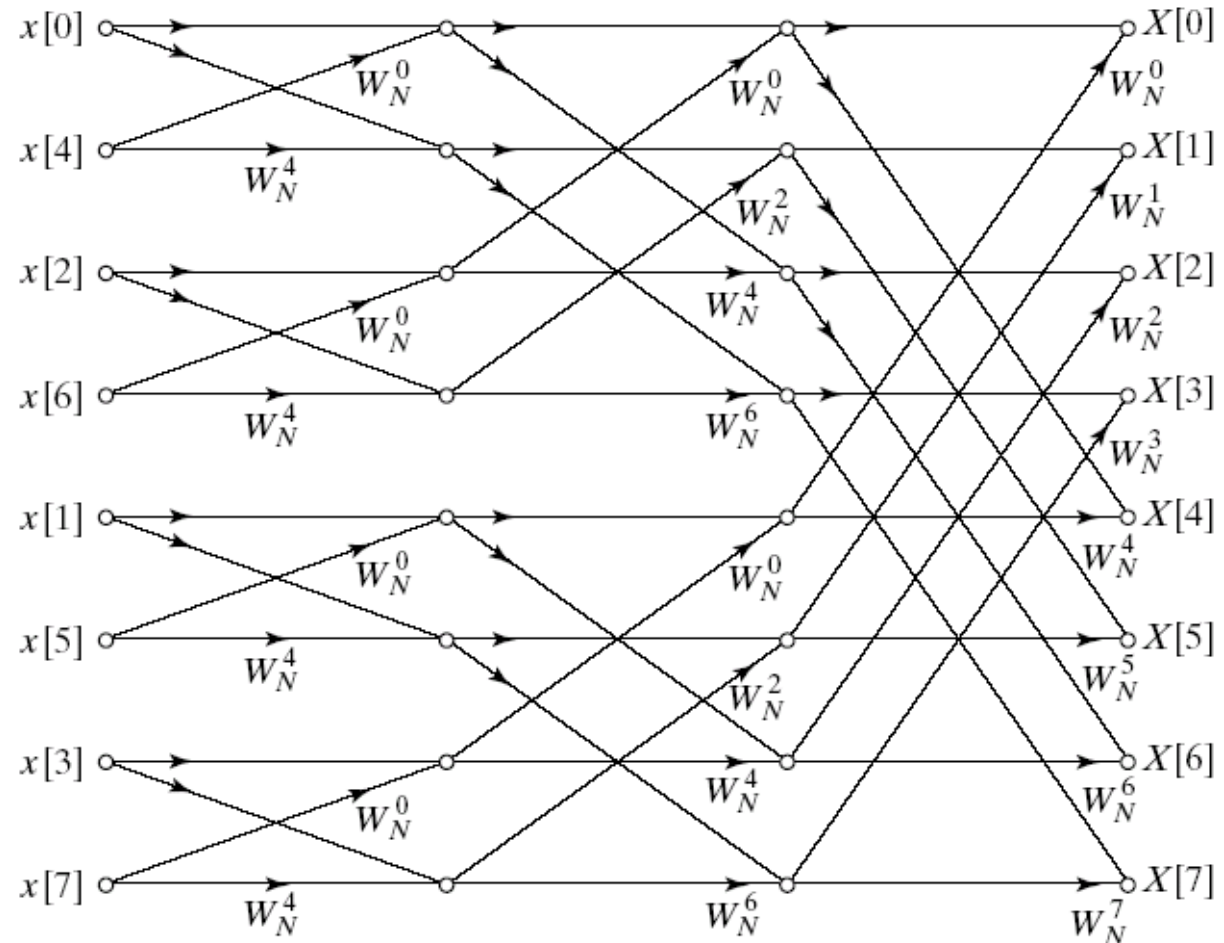
- After two steps of decimation in



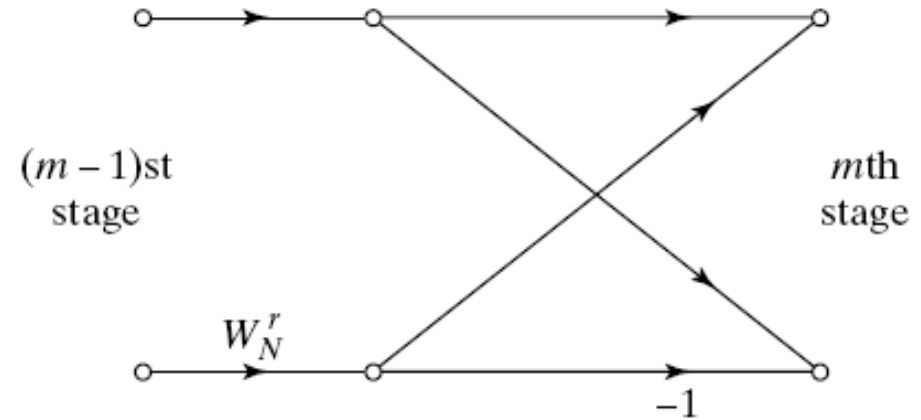
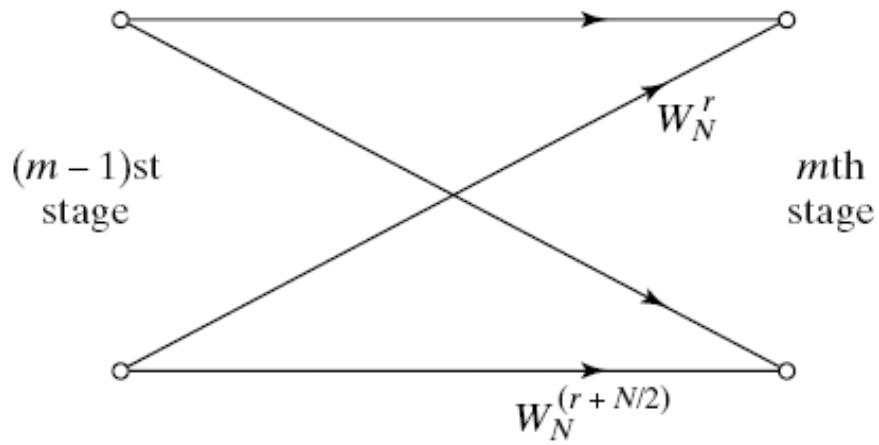
- Repeat until we're left with two-point DFT's



- Flow graph for 8-point decimation in time

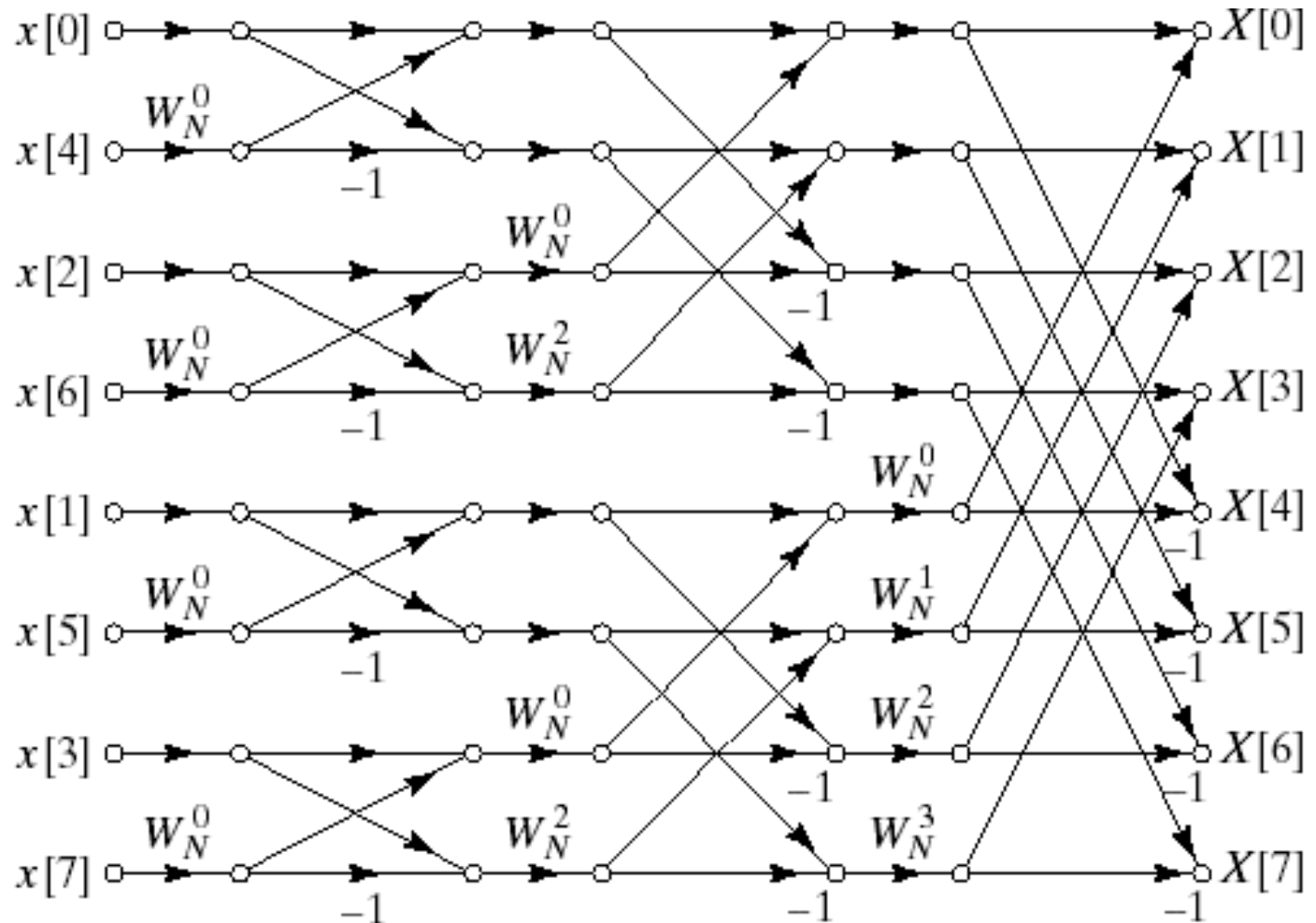


- Flow graph constitutes of butterflies



- Final flow graph for 8-point decimation in time

$(N \log_2 N)/2$
complex
Multiplications
and
 $N \log_2 N$
additions

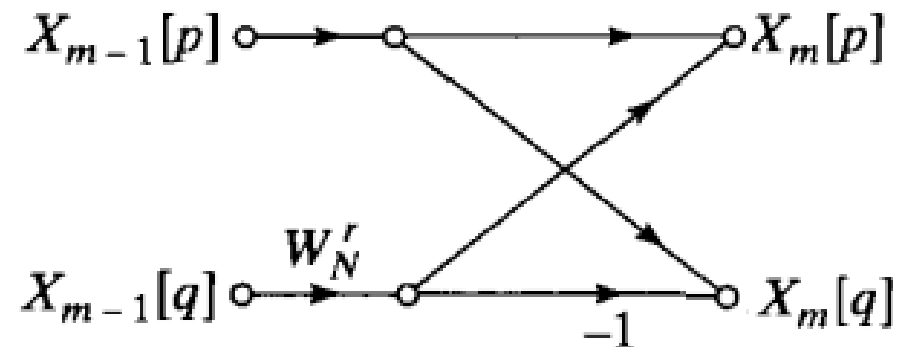


Computation of DFT

- Decimation-in-time flow graphs require two sets of registers
- Input and output for each stage

$$X_m[p] = X_{m-1}[p] + W_N^r X_{m-1}[q]$$

$$X_m[q] = X_{m-1}[p] - W_N^r X_{m-1}[q]$$



$$X_0[0] = x[0]$$

$$X_0[1] = x[4]$$

$$X_0[2] = x[2]$$

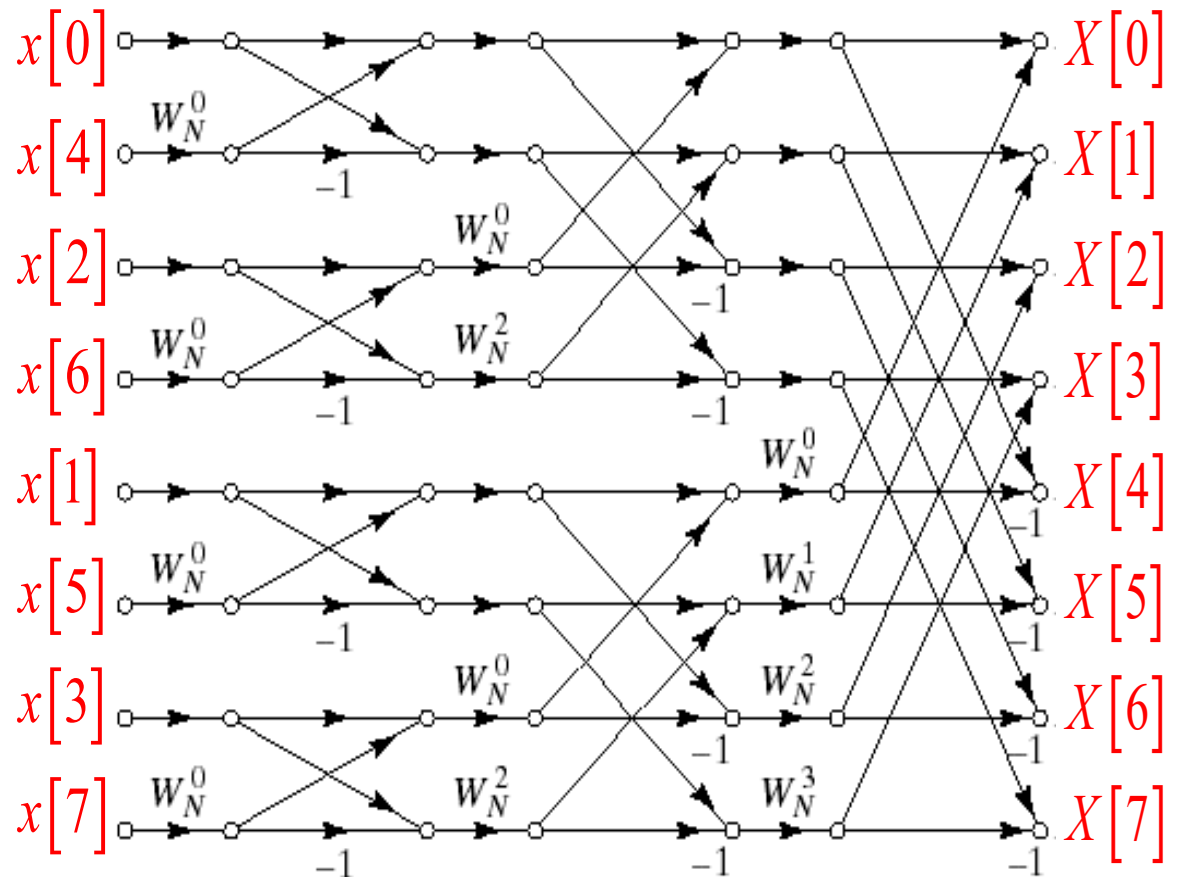
$$X_0[3] = x[6]$$

$$X_0[4] = x[1]$$

$$X_0[5] = x[5]$$

$$X_0[6] = x[3]$$

$$X_0[7] = x[7]$$



1. Decimation-in-time FFT Algorithms

2. Decimation-in-frequency FFT Algorithms

- The DFT equation

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}$$

- Split the DFT equation into **even** and **odd** frequency indexes

$$X[2r] = \sum_{n=0}^{N-1} x[n] W_N^{n2r} = \sum_{n=0}^{N/2-1} x[n] W_N^{n2r} + \sum_{n=N/2}^{N-1} x[n] W_N^{n2r}$$

- Substitute variables

$$\begin{aligned} &= \sum_{n=0}^{N/2-1} x[n] W_N^{n2r} + \sum_{n=0}^{N/2-1} x[n + N/2] W_N^{(n+N/2)2r} \\ &= \sum_{n=0}^{N/2-1} (x[n] + x[n + N/2]) W_{N/2}^{nr} = \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{nr} \end{aligned}$$

- The DFT equation

$$X[k] = \sum_{n=0}^{N-1} x[n] W_N^{nk}$$

$$X[2r+1] = \sum_{n=0}^{N-1} x[n] W_N^{n(2r+1)} = \sum_{n=0}^{N/2-1} x[n] W_N^{n(2r+1)} + \sum_{n=N/2}^{N-1} x[n] W_N^{n(2r+1)}$$

$$= \sum_{n=0}^{N/2-1} x[n] W_N^{n(2r+1)} + \sum_{n=0}^{N/2-1} x[n + N/2] W_N^{(n+N/2)(2r+1)}$$

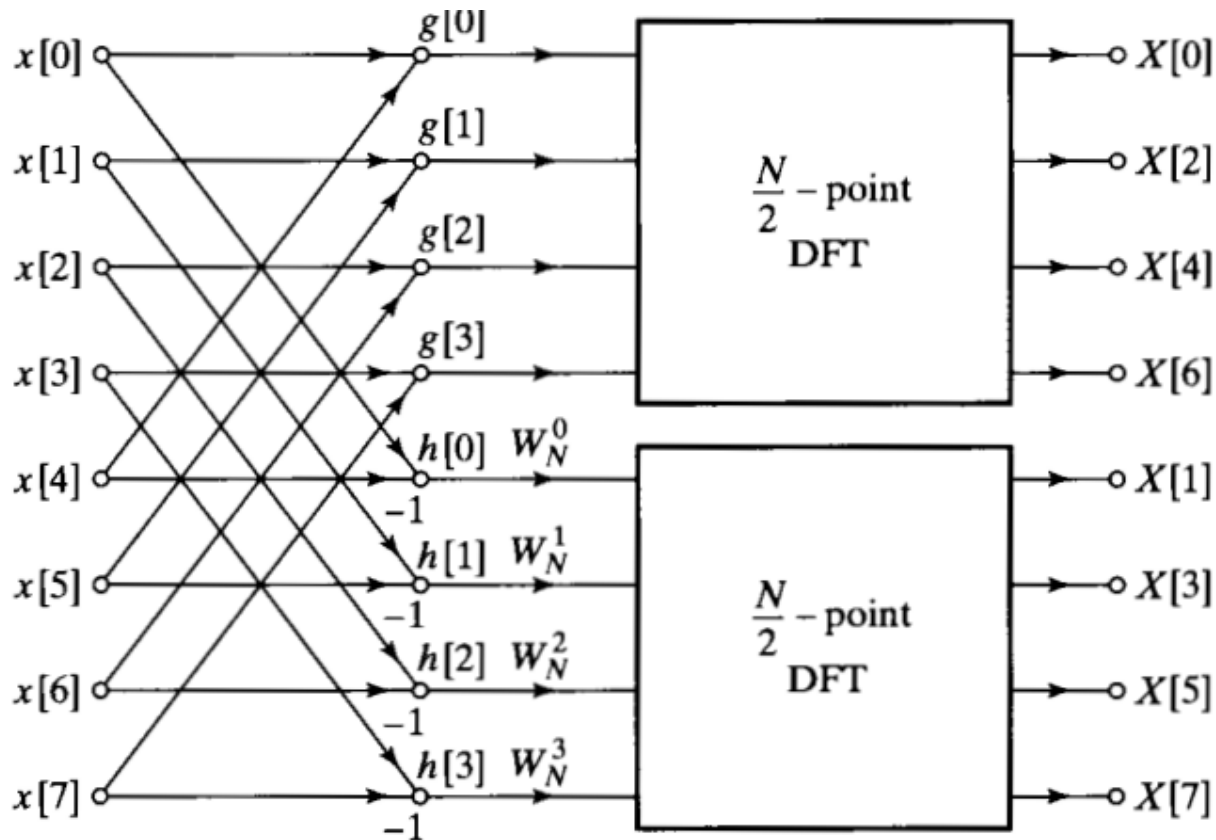
$$= \sum_{n=0}^{N/2-1} (x[n] - x[n + N/2]) W_N^{n(2r+1)} = \sum_{n=0}^{N/2-1} (x[n] - x[n + N/2]) W_N^n W_{N/2}^{rn}$$

$$= \sum_{n=0}^{N/2-1} h(n) W_N^n W_{N/2}^{rn}$$

$$W_N^{n(2r+1)} = W_N^{2rn} W_N^n = W_{N/2}^{rn} W_N^n$$

$$W_N^{\frac{N}{2}(2r+1)} = W_N^{Nr} W_N^{N/2} = -1$$

- Decimation-in-frequency decomposition of an N-point DFT to N/2-point DFT

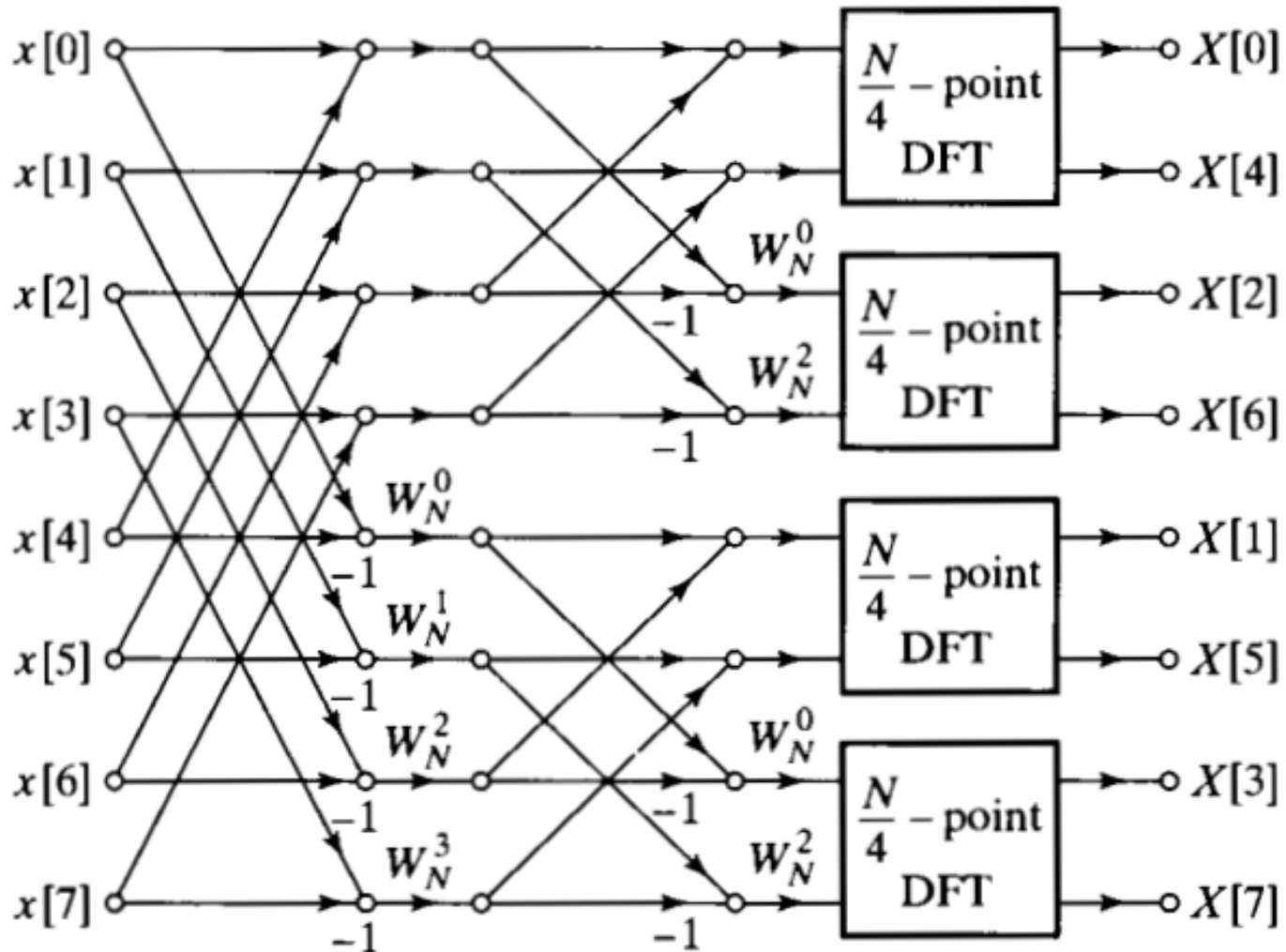


$$X[2r] = \sum_{n=0}^{N/2-1} (x[n] + x[n + N/2]) W_{N/2}^{nr}$$

$$= \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{rn}$$

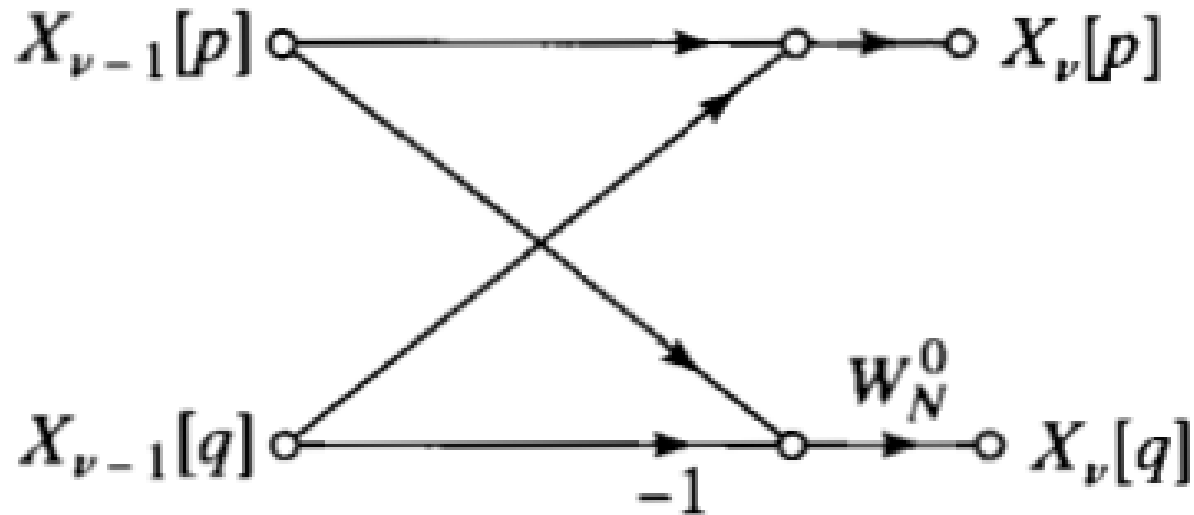
$$X[2r+1] = \sum_{n=0}^{N/2-1} (x[n] - x[n + N/2]) W_N^n W_{N/2}^{rn}$$

$$= \sum_{n=0}^{N/2-1} h(n) W_N^n W_{N/2}^{rn}$$



$$\begin{aligned}
 X[2 * 2s] &= \sum_{n=0}^{N/4-1} [g(n) + g(n + N / 4)] W_{N/4}^{sn} \\
 &= \sum_{n=0}^{N/4-1} p(n) W_{N/4}^{sn} \\
 X[2 * (2s + 1)] &= \sum_{n=0}^{N/4-1} [g(n) - g(n + N / 4)] W_N^{2n} W_{N/4}^{sn} \\
 &= \sum_{n=0}^{N/4-1} q(n) W_N^{2n} W_{N/4}^{sn}
 \end{aligned}$$

- 2-point DFT



$$X_v(p) = X_{v-1}(p) + X_{v-1}(q)$$

$$X_v(q) = [X_{v-1}(p) - X_{v-1}(q)] W_8^0 \quad \text{when } N = 8$$

$$\begin{aligned}
 X[2r] &= \sum_{n=0}^{N/2-1} (x[n] + x[n + N/2]) W_{N/2}^{nr} = \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{rn} \\
 X[2 * 2s] &= \sum_{n=0}^{N/4-1} g(n) W_{N/2}^{2sn} + \sum_{n=N/4}^{N/2-1} g(n) W_{N/2}^{2sn} \\
 &= \sum_{n=0}^{N/4-1} g(n) W_{N/2}^{2sn} + \sum_{n=0}^{N/4-1} g(n + N/4) W_{N/2}^{2s(n+N/4)} \\
 &= \sum_{n=0}^{N/4-1} g(n) W_{N/4}^{sn} + \sum_{n=0}^{N/4-1} g(n + N/4) W_{N/4}^{sn} \\
 &= \sum_{n=0}^{N/4-1} [g(n) + g(n + N/4)] W_{N/4}^{sn} = \sum_{n=0}^{N/4-1} p(n) W_{N/4}^{sn}
 \end{aligned}$$

$$X[2r] = \sum_{n=0}^{N/2-1} (x[n] + x[n + N/2]) W_{N/2}^{nr} = \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{rn}$$

$$X[2*(2s+1)] = \sum_{n=0}^{N/2-1} g(n) W_{N/2}^{(2s+1)n}$$

$$= \sum_{n=0}^{N/4-1} g(n) W_{N/2}^{(2s+1)n} + \sum_{n=N/4}^{N/2-1} g(n) W_{N/2}^{(2s+1)n}$$

$$= \sum_{n=0}^{N/4-1} g(n) W_{N/4}^{sn} W_{N/2}^n + \sum_{n=0}^{N/4-1} g(n + N/4) W_{N/2}^{(2s+1)(n+N/4)}$$

$$= \sum_{n=0}^{N/4-1} g(n) W_{N/4}^{sn} W_N^{2n} + \sum_{n=0}^{N/4-1} g(n + N/4) W_{N/4}^{sn} W_N^{2n} W_{N/2}^{(2s+1)N/4}$$

$$= \sum_{n=0}^{N/4-1} [g(n) - g(n + N/4)] W_N^{2n} W_{N/4}^{sn}$$

$$W_{N/2}^{(2s+1)N/4} = W_{N/2}^{sN/2} W_{N/2}^{N/4} = -1$$

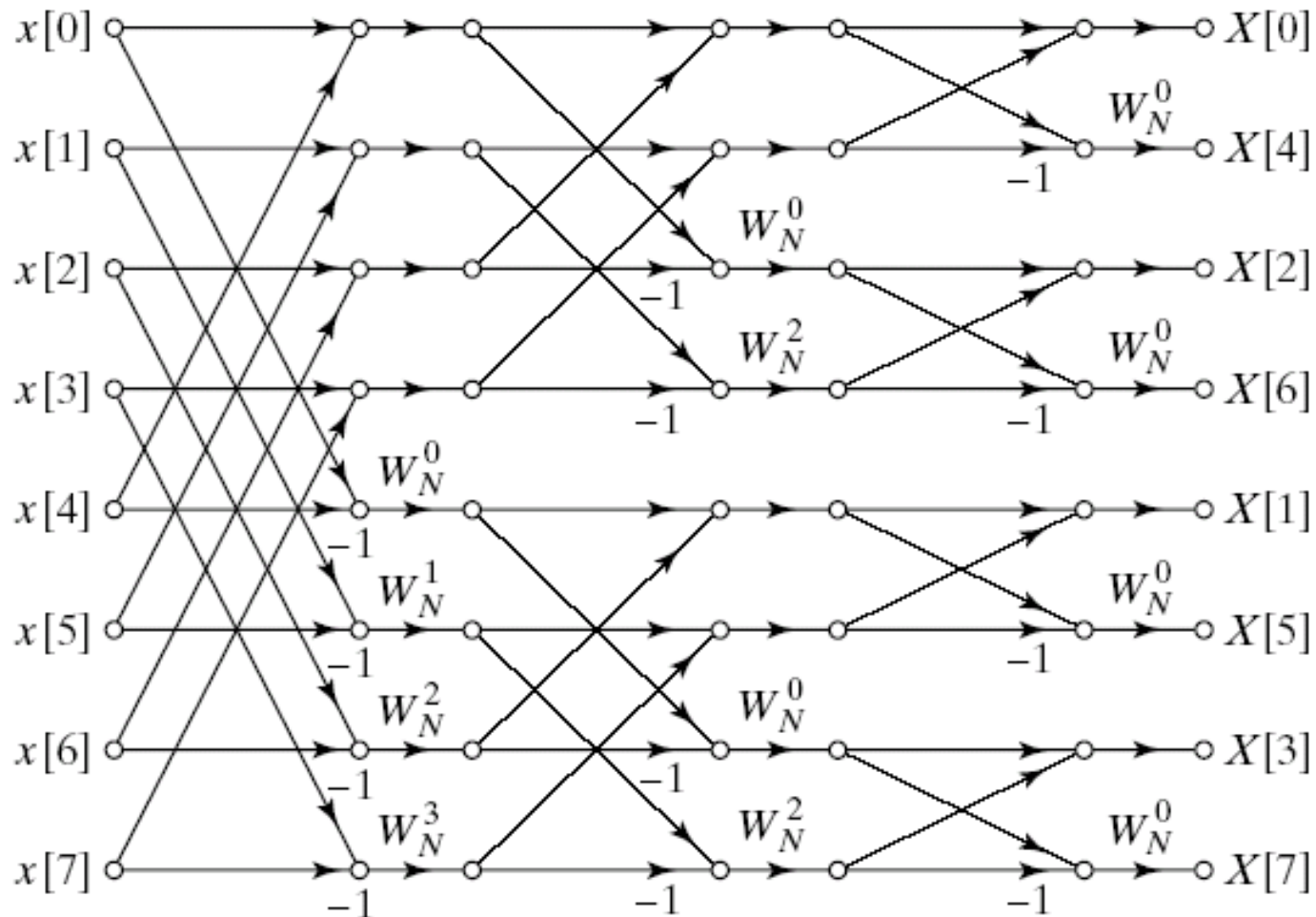
$$\begin{aligned}
 X[2 * 2 * 2t] &= \sum_{n=0}^{N/4-1} p(n) W_{N/4}^{2tn} & X[2 * 2s] &= \sum_{n=0}^{N/4-1} p(n) W_{N/4}^{sn} \\
 &= \sum_{n=0}^{N/8-1} p(n) W_{N/4}^{2tn} + \sum_{n=N/8}^{N/4-1} p(n) W_{N/4}^{2tn} \\
 &= \sum_{n=0}^{N/8-1} p(n) W_{N/4}^{2tn} + \sum_{n=0}^{N/8-1} p(n + N/8) W_{N/4}^{2t(n+N/8)} \\
 &= \sum_{n=0}^{N/8-1} [p(n) + p(n + N/8)] W_{N/8}^{tn} = p(n) + p(n+1) \text{ when } N = 8
 \end{aligned}$$

$$\begin{aligned}
 X[2 * 2 * (2t + 1)] &= \sum_{n=0}^{N/4-1} p(n) W_{N/4}^{(2t+1)n} & X[2 * 2s] &= \sum_{n=0}^{N/4-1} p(n) W_{N/4}^{sn} \\
 &= \sum_{n=0}^{N/8-1} p(n) W_{N/4}^{(2t+1)n} + \sum_{n=N/8}^{N/4-1} p(n) W_{N/4}^{(2t+1)n} \\
 &= \sum_{n=0}^{N/8-1} p(n) W_{N/4}^{(2t+1)n} + \sum_{n=0}^{N/8-1} p(n + N/8) W_{N/4}^{(2t+1)(n+N/8)} \\
 &= \sum_{n=0}^{N/8-1} p(n) W_{N/4}^{2tn} W_{N/4}^n + \sum_{n=0}^{N/8-1} p(n + N/8) W_{N/4}^{2tn} W_{N/4}^n W_{N/4}^{(2t+1)N/8} \\
 &= \sum_{n=0}^{N/8-1} [p(n) - p(n + N/8)] W_{N/8}^{tn} W_N^{4n} = [p(n) - p(n+1)] W_8^0
 \end{aligned}$$

when $N = 8$

$$W_{N/4}^{(2t+1)N/8} = W_{N/4}^{tN/4} W_{N/4}^{N/8} = -1$$

• Final flow graph for 8-point DFT decimation in frequency





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THANK YOU